A Companion to Shamkanov's *Circular Proofs for the Gödel–Löb Provability Logic*

Understanding how Proof System Transformations, Circularity, and Fixed-point Theorems, give Interpolation in GL

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Abstract

This is a companion to the 2014 paper by D. S. Shamkanov [1], which gives the first syntactic proof of the Lyndon interpolation property for the Gödel–Löb provability logic **GL**. Instead of an explanation of his whole proof flush with the many theorems and lemmas, this companion takes on, for elaboration and explanation, the most important questions that naturally arise when trying to follow Shamkanov's paper: what is it about the standard modal rule for $\mathbf{GL_{Seq}}$ that makes Lyndon interpolation difficult; why replacing it with a simpler rule in $\mathbf{GL}_\infty$ makes infinite proofs possible; how the admissibility of the Löb rule and the structure of circular proofs together give the three-way equivalence; and how the GL fixed-point theorem resolves the twofold self-reference encountered when building interpolants from split circular proofs.

Note that all theorem numbers, lemma numbers, and equation labels match those of [1] unless otherwise indicated.

1. Background: Provability Logic and Why It Matters

Gödel's 1931 incompleteness theorems revealed that Peano Arithmetic (**PA**) contains a predicate $\mathrm{Prov}(x)$,definable in the language of arithmetic, that expresses "the formula with code x is provable in **PA**." This predicate satisfies three *derivability conditions* (Hilbert–Bernays–Löb): it distributes over implication, is self-aware (if $\mathbf{PA} \vdash \varphi$ then $\mathbf{PA} \vdash \mathrm{Prov}(\ulcorner\varphi\urcorner)$), and is introspective ($\mathrm{Prov}(\ulcorner\varphi\urcorner) \to \mathrm{Prov}(\ulcorner\mathrm{Prov}(\ulcorner\varphi\urcorner)\urcorner)$ is provable). Logician Löb wondered then, if it is possible to characterize formulae for which it is possible to prove their provability will give their truth. This resulted in his coming up with Löb's theorem [3]:

Theorem 1.1 (Löb, 1955)**.** $\mathbf{PA} \vdash \mathrm{Prov}(\ulcorner\varphi\urcorner) \to \varphi$ *if and only if* $\mathbf{PA} \vdash \varphi$.

Formulae for which their provability gives truth are those that are already provable outright.

It was soon noticed that replacing $\mathrm{Prov}(\ulcorner\varphi\urcorner)$ by a modal operator $\Box\varphi$ turns the derivability conditions into the axioms of a modal logic. The *Gödel–Löb logic* **GL** is the normal modal logic axiomatised by the classical tautologies, $\Box(A \to B) \to (\Box A \to \Box B)$, and *Löb's axiom*

$$\Box(\Box A \to A) \to \Box A.$$

By Solovay's completeness theorem [4], **GL** is exactly the logic of arithmetic provability: a modal formula is provable in **GL** if and only if all its arithmetic substitution instances are provable in **PA**. Semantically, **GL** is sound and complete for the class of *finite strict partial orders* (Segerberg [5]): Kripke frames in which the accessibility relation is transitive, irreflexive, and well-founded. The well-foundedness is the semantic counterpart of Löb's theorem; this turns out to be a crucial point when thinking of generalisations of Shamkanov's proof beyond **GL**.

The paper [1] works in the one-sided Tait-style sequent calculus $\mathbf{GL_{Seq}}$, in which a sequent Γ is a finite multiset of formulas whose intended meaning is $\Gamma^\sharp = \bigvee_{A\in\Gamma} A$. Formulas are built from propositional constants, their duals $\bar{A}$ (defined via de Morgan's laws so that $\overline{\Box A} = \Diamond\bar{A}$ and $\bar{\bar{A}} = A$), and the connectives $\wedge, \vee, \Box, \Diamond$. The calculus $\mathbf{GL_{Seq}}$ has axioms $\Gamma, A, \bar{A}$ and $\Gamma, \top$, propositional rules $\wedge, \vee$, and the modal rule $\Box_{\mathbf{GL}}$ (recalled below).

2. The $\Box_{\mathbf{GL}}$ Rule and the Obstacle to Lyndon Interpolation

2.1. The rule and its meaning

The modal rule of $\mathbf{GL_{Seq}}$ is:

$$\frac{\Gamma,\ \Diamond\Gamma,\ \Diamond\bar{A},\ A}{\Diamond\Gamma,\ \Box A,\ \Delta}\ [\Box_{\mathbf{GL}}] \tag{1}$$

From the bottom-up: to justify $\Box A$ in the conclusion $\Diamond\Gamma, \Box A, \Delta$, It is needed that A is proved in a context consisting of:

- Γ : the raw background context;
- $\Diamond\Gamma$: the holding of transitivity,i.e, in the accessible worlds the same background holds (since $\mathbf{GL}$ has the transitivity axiom $\Box A \to \Box\Box A$);
- $\Diamond\bar{A}$: the *Löb challenge*: some accessible world may falsify A.

There is particular emphasis on $\Diamond\bar{A}$ as the term encodes Löb's theorem directly: one must prove A even in the face of a potential counterexample, preventing the circular inference "A is provable, therefore A." It is useful to notice, and keep in mind, that even when moving on to circular proofs, the structure of this logic ensures the exclusion of outright circular reasoning.

2.2. Why this breaks Lyndon interpolation

The *Lyndon interpolation property* for $\mathbf{GL}$ states: if $\mathbf{GL} \vdash A \to B$ then there exists C with $w(C) \subseteq w(A) \cap w(B)$, $\mathbf{GL} \vdash A \to C$, and $\mathbf{GL} \vdash C \to B$, where $w(\cdot)$ records both the polarity and the modal position (inside or outside $\Box$) of each literal.

The standard syntactic method for interpolation i.e, the *Maehara method*, works by labelling every formula in a cut-free proof as belonging to the "A-side" or the "B-side," and propagating these labels upward through each inference rule. Of course, this onely works without hassle when formulas maintain a consistent polarity role as one works their way up the proof.

It is not hard at all then, to see that rule (1) breaks this. The formula A appears positively in the conclusion (inside $\Box A$) and in the premise (directly, as A); but $\bar{A}$ also appears in the premise (as $\Diamond\bar{A}$). Assigning $\Box A$ to one side forces A and $\bar{A}$ to appear simultaneously in the premise on both sides, destroying any coherence in polarity preservation and bookeeping. Hence, despite $\mathbf{GL}$ being known to have the Lyndon interpolation property semantically [2], no syntactic proof existed before [1].

TL;DR: The $\Diamond\bar{A}$ term in the premise of $\Box_{\mathbf{GL}}$ is a syntactic encoding of Löb's theorem. It makes the rule semantically powerful but very hard to work with from a proof theoretic perspective: any formula assigned to the left side of an interpolating split also appears on the right negated, making a consistent polarity assignment impossible.

3. Replacing $\Box_{\mathbf{GL}}$ with a Simpler Rule: The Price of Infinity

The remedy? Replace $\Box_{\mathbf{GL}}$ with the *K4 modal rule*:

$$\frac{\Gamma,\ \Diamond\Gamma,\ A}{\Diamond\Gamma,\ \Box A,\ \Delta}\ [\Box] \tag{2}$$

Rule (2) has no $\Diamond\bar{A}$ in the premise and therefore makes a proper polarity assignment possible: A appears once, and it can be assigned to one side of any split. This restores the applicability of the Maehara method.

However, rule (2) is strictly weaker than $\Box_{\mathbf{GL}}$: meaning it cannot prove Löb's axiom by itself. There is a missing piece that must come from somewhere.

3.1. Where the infinity comes from

The three-way equivalence $\mathbf{GL_{Seq}} \equiv \mathbf{GL}_\infty \equiv \mathbf{GL}_{\text{circ}}$ (Theorem 3.2 of [1]) is proved in one direction by Lemma 3.3: given a finite $\mathbf{GL_{Seq}}$-proof π, construct a (possibly infinite) $\mathbf{GL}_\infty$-proof. The construction replaces each application of $\Box_{\mathbf{GL}}$ with simpler rule (2), but first must be drop the $\Diamond\bar{A}$ from the premise. This is taken care of by applying the admissible *Löb rule* (Proposition 2.1 of [1]):

$$\frac{\Gamma, \Diamond\Gamma, \Diamond\bar{A}, A}{\Gamma, \Diamond\Gamma, A}\ [\text{Löb}]$$

The Löb rule is admissible in $\mathbf{GL_{Seq}}$ (though this is not entirely trivial with the popping of infinity, hence leading to some confusion): whenever its premise is provable, so is its conclusion. Applying it to the premise τ of a $\Box_{\mathbf{GL}}$ step produces a new proof $f(\tau)$ of $\Gamma, \Diamond\Gamma, A$. This transformation h then applies itself recursively to $f(\tau)$ as the new proof itself may contain the $\Box_{\mathbf{GL}}$ rule, needing a new proof from its premises to conclusion and so on.

The key observation to note here is that the admissibility proof for the Löb rule uses $\Box_{\mathbf{GL}}$ applied to τ, embedding τ inside a larger proof. So $f(\tau)$ contains all the $\Box_{\mathbf{GL}}$ rules that τ contained, plus at least one new one. When h processes $f(\tau)$, it again meets $\Box_{\mathbf{GL}}$ rules, applies the Löb rule again, producing proofs of yet greater complexity, and so on without bound. The output is possibly an infinite tree.

Also, observe that the transformation h is defined by *corecursion*: (not by induction on some decreasing measure), but as the unique fixed point of a contractive operator F on the complete metric space $\mathcal{C}$ of all "colourings" of the infinite binary tree, and a metric defined by $d(u, v) = 2^{-\delta(u,v)}$ where $\delta(u, v)$ is the depth of the shallowest node at which u and v differ. Since $F(h)(c)$ depends on h only at immediate subtrees of c (one level deeper), F is a contraction with factor $\frac{1}{2}$, and the Banach fixed-point theorem guarantees a unique h [1, Sec. 3].

TL;DR: Switching from $\Box_{\mathbf{GL}}$ to the simpler $\Box$ removes the polarity obstacle by discarding the Löb portion of the rule. Lemma 3.3 recovers that via the admissible Löb rule, at the cost of producing a proof that may be infinite. Infinite proofs are legitimate in $\mathbf{GL}_\infty$; Lemma 3.6 then compresses them back into finite circular proofs via the subformula property and a pigeonhole argument.

4. Admissibility of the Löb Rule, Circularity, and the Proof of $\mathbf{GL}_{\text{circ}} \Rightarrow \mathbf{GL_{Seq}}$

The most interesting, but confusing part of Theorem 3.2 is Lemma 3.7: every sequent provable in $\mathbf{GL}_{\text{circ}}$ is provable in $\mathbf{GL_{Seq}}$. This is the direction that shows circular proofs are sound i.e, they do not prove anything illegitimate as circular argumentation is usually associated with.

4.1. The main claim and its induction

Let $\pi = (\kappa, d)$ be a circular derivation of Γ. Note: $H(\pi)$ denotes the non-axiom leaves of κ that have no back-link and no application of $\Box$ on the path from the leaf to the root, and $BH(\pi)$ denotes those with no back-link but with at least one $\Box$ on that path. $\overline{\Box} A = A \wedge \Box A$ is the notation used.

The claim proved by induction on the structure of κ is:

$$\mathbf{GL} \vdash \Big(\bigwedge_{a \in H(\pi)} \overline{\Box}\, \Delta_a^\sharp \Big) \wedge \Big(\bigwedge_{a \in BH(\pi)} \Box \Delta_a^\sharp \Big) \longrightarrow \Gamma^\sharp. \tag{3}$$

The hypothesis says that all open-assumption sequents hold, with with respect to their modal positioning, that is, a leaf in H contributes both the actual truth $\Delta_a^\sharp$ and its provability $\Box\Delta_a^\sharp$; a leaf in BH contributes only $\Box\Delta_a^\sharp$, because the intervening $\Box$ rule strips away the truth from the conclusion. For a complete circular proof $H(\pi) = BH(\pi) = \emptyset$, so (3) giving $\mathbf{GL} \vdash \Gamma^\sharp$ directly.

The base cases and propositional rule cases are taken care of without hassle. The $\Box$ rule case transfers H-assumptions into BH-assumptions (they become "boxed" after crossing the $\Box$ rule) and uses the GL-valid $\Box(\Gamma, \Diamond\Gamma, A)^\sharp \to (\Diamond\Gamma, \Box A, \Sigma)^\sharp$.

4.2. The back-link case and the Löb rule

The critical case in the case distinction is when some leaf b is connected by a back-link to the root, so $\Delta_b = \Gamma$. The argument has three steps.

Step 1. Erase the back-link to obtain π_0. Now b is an open assumption. Since the back-link must pass through a $\Box$ rule (applies for all back links according to Definition 3.3 in [1]), $b \in BH(\pi_0)$.

Step 2. Apply the induction hypothesis to π_0; which has one fewer back-link making $BH(\pi_0) = BH(\pi) \cup \{b\}$ and $\Delta_b = \Gamma$, the conclusion is:

$$\mathbf{GL} \vdash (\text{assumptions of } \pi) \wedge \underbrace{\Box\Gamma^\sharp}_{\text{from } b} \to \Gamma^\sharp.$$

Step 3. The formula $\Box\Gamma^\sharp \to \Gamma^\sharp$ is exactly a Löb-hypothesis type pattern. The admissible Löb in $\mathbf{GL_{Seq}}$ means $\Box\Gamma^\sharp$ can be dropped, yielding $\mathbf{GL} \vdash (\text{assumptions of } \pi) \to \Gamma^\sharp$, which is exactly (3) for π.

TL;DR: What the Löb rule is doing. The back-link says: "I will prove $\Gamma^\sharp$ by assuming it is already provable." This is a circular provability assumption, which is precisely the pattern that Löb's theorem is designed to discharge. Löb's axiom encodes the well-foundedness of provability in **PA**: there are no infinite relapse of the form "provably provably provably…". The admissible Löb rule makes this well-foundedness available at the syntactic level. Correspondingly, *back-links of the form in quesiton are Löb's theorem structural form*: the Löb principle is transfered from the $\Box_{\mathbf{GL}}$ inference rule in $\mathbf{GL_{Seq}}$ into the back-link structure of $\mathbf{GL}_{\text{circ}}$, and the admissible Löb rule resolves the circulairty when going the other way around.

5. Twofold Self-Reference in Interpolant Construction and Its Resolution

In Section 4 of [1], Theorem 4.1 (Lyndon interpolation) is proved. The structure of the argument follows the same back-link pattern as Lemma 3.7, but with an additional level of self-reference that requires a second tool to come to the rescue.

5.1. The background: Interpolant construction from split circular proofs

A *splitting* $\Gamma_1 \mid \Gamma_2$ of a provable sequent Γ_1, Γ_2 assigns each formula to a left or right side. An *interpolant of* $\Gamma_1 \mid \Gamma_2$ is a formula C with $w(C) \subseteq w(\bar{\Gamma}_1) \cap w(\Gamma_2)$, $\mathbf{GL} \vdash \bar{\Gamma}_1^\sharp \to C$, and $\mathbf{GL} \vdash C \to \Gamma_2^\sharp$. By Lemma 4.2 of [1], every circular proof of Γ_1, Γ_2 yields a *split circular proof* of $\Gamma_1 \mid \Gamma_2$.

The *split inference rules* (equation (1) in [1]) tell us, for each rule of $\mathbf{GL}_{\text{circ}}$, how the split propagates from conclusion to premises and what the interpolant at the conclusion is in terms of the interpolants at the premises. For propositional rules this is once again dealt with rather smoothly. For the modal rule there are two cases depending on which side $\Box A$ lands:

- $\Box A$ on the *left*: premise has A on the left; interpolant becomes $\Diamond C$.
- $\Box A$ on the *right*: premise has A on the right; interpolant becomes $\Box C$.

These are exactly the cases where the K4 $\Box$ rule (rule (2)) comes in handy: since A appears only once in the premise, it can be assigned unambiguously to one side.

5.2. The twofold circularity

The inductive construction assigns concrete interpolants to all non-bud nodes. The obstruction arises, rather unsurprisingly, at the back-links. Suppose leaf b is connected to the root, meaning they carry the same split sequent $\Gamma_1 \mid \Gamma_2$.

Just like before, erase the back-link to get π_0; now b is an open assumption. Introduce a placeholder variable X_b for the (unknown) interpolant at b. Proceeding inductively through π_0 yields a formula C_0, possibly containing X_b, satisfying (for suitable assumptions of π_0):

$$\mathbf{GL} \vdash (\text{assumptions}) \wedge \Box I_b \;\rightarrow\; \big((\bar{\Gamma}_1^\sharp \rightarrow C_0) \wedge (C_0 \rightarrow \Gamma_2^\sharp)\big), \tag{4}$$

where $I_b = (\bar{\Gamma}_1^\sharp \rightarrow X_b) \wedge (X_b \rightarrow \Gamma_2^\sharp)$ is the statement that X_b is itself an interpolant for $\Gamma_1 \mid \Gamma_2$.

It is possible to observe that two levels of self-reference are now present:

(i) **Formula level.** The variable X_b appears in C_0, and X_b is supposed to equal the interpolant at the root: which is C_0 itself! (since b and the root carry the same sequent). So C_0 contains itself as a component.

(ii) **Provability level.** The hypothesis $\Box I_b$ in (4) asserting the *provability* of "X_b is an interpolant" : a circular provability assumption of the same kind as $\Box\Gamma^\sharp$ in Lemma 3.7.

5.3. Resolution by the fixed-point theorem and the Löb rule

Level (i): the GL fixed-point theorem. Since the back-link passes through a $\Box$ rule, the variable X_b appears in C_0 only inside modal operators. Upon this observation, Theorem 4.3 of [1] (the GL fixed-point theorem) then comes to one's rescue: for any formula $A(P)$ in which P occurs only under $\Box$, there exists a P-free formula F with $w(F) \subseteq w^*(A) \setminus \{P^\circ\}$ and $\mathbf{GL} \vdash \overline{\Box}(P \leftrightarrow A) \leftrightarrow \overline{\Box}(P \leftrightarrow F)$. Applying this to $X_b \leftrightarrow C_0(X_b)$ gives a concrete formula F eliminating X_b. Obtained by setting $C := C_0[X_b \mapsto F]$.

The GL fixed-point theorem can be used to resolve this circulairty because **GL**'s Löb axiom makes any "modalized" self-referential equation well-founded: the self-reference is always shielded by $\Box$, so it concerns provability of X_b rather than X_b's truth directly, and this well-foundedness of provability guarantees a unique solution.

Level (ii): the Löb rule. After substituting F for X_b, condition (4) becomes:

$$\mathbf{GL} \vdash (\text{assumptions of } \pi) \wedge \Box I_b(F) \;\rightarrow\; \big((\bar{\Gamma}_1^\sharp \rightarrow C) \wedge (C \rightarrow \Gamma_2^\sharp)\big).$$

Now $I_b(F)$ is exactly the statement that C is an interpolant for the root split sequent. So the hypothesis $\Box I_b(F)$ asserts the provability of the very conclusion we are trying to establish, a Löb-pattern circular provability assumption. The admissible Löb rule drops it, giving:

$$\mathbf{GL} \vdash (\text{assumptions of } \pi) \;\rightarrow\; \big((\bar{\Gamma}_1^\sharp \rightarrow C) \wedge (C \rightarrow \Gamma_2^\sharp)\big).$$

For a complete split circular proof (withno open assumptions), C is the desired Lyndon interpolant.

TL;DR: Why two tools are needed. In Lemma 3.7 only the Löb rule was needed because the induction tracked the sequent $\Gamma^\sharp$ directly: the back-link created one level of self-reference (the circular provability assumption $\Box\Gamma^\sharp$), which the Löb rule dissolved. In Theorem 4.1 the induction tracks a *formula to be constructed*, i.e, the interpolant. The back-link creates a self-referential formula placeholder X_b in addition to a circular provability assumption. The

placeholder must be eliminated first, by the fixed-point theorem, which is applicable because the placeholder X_b only appears inside the modal operator in C_0, allowing for its elimination, there-by resolving the first self reference. Then the usual argumentation from before applies.

Concluding Remarks: From GL to the Modal μ-Calculus

The four difficulties addressed in this companion are not independent, nor are they merely to be understood as quirks. They form the core of the proof and point us toward a possible generalization: the $\Diamond\bar{A}$ in $\Box_{\mathbf{GL}}$ encodes Löb's theorem inside an inference rule; removing it forces infinite proofs; those infinite proofs compress into circular proofs because of the subformula property; circular proofs are sound precisely because back-links generate Löb-pattern hypotheses that the admissible Löb rule can discharge; and the same discharge, lifted one level to the interpolant formula rather than the sequent, requires the additional mediation of the GL fixed-point theorem. Shamkanov's paper, in some sense, almost hints that the self-referential content of **GL** is best handled by proof-theoretic tools that are themselves self-referential.

What makes these observations quite interesting is the fact that each feature of Shamkanov's argument can be understood as what is possibly an instance of a general set of conditions for such proofs and tools, a notion that enables the generalisation carried out by Afshari and Leigh [**?**, **?**].

Well-foundedness becomes the global trace condition. In **GL**, the requirement that every back-link pass through a $\Box$ rule is a local, checkable condition, justified by the well-foundedness of provability in **PA**. In the μ-calculus, where one reasons about arbitrary inductive and coinductive definitions, this localises into a global condition on infinite paths: a cyclic proof is sound when every infinite path carries a *good trace*, meaning the formula thread along that path eventually stabilises on a μ-formula (for a left trace) or a ν-formula (for a right trace). The admissible Löb rule, which discharges circular provability assumptions one back-link at a time, is replaced by the *Normal Form Theorem*, which rearranges any cyclic proof so that a global marking σ assigns μ or ν to each companion consistently with the trace condition. This reinforces the notion that some well-foundedness condition may always be necessary when dealing with circular proof systems for a logic.

The fixed-point theorem becomes an internal quantifier. In Shamkanov's proof, the self-referential placeholder X_b in the interpolant formula is eliminated by invoking the GL fixed-point theorem as a tool: the modalized equation $X_b \leftrightarrow C_0(X_b)$ has a canonical solution F, owing to Löb's axiom, and one sets $C := C_0[X_b \mapsto F]$. In the μ-calculus, this step disappears as a separate argument because the language already contains fixed-point quantifiers: the interpolant at a companion is simply $\mu x_b\, C_0(x_b)$ or $\nu x_b\, C_0(x_b)$, with the choice of quantifier given by the marking σ. This emphasises the necessity of some fix-point properties in logics where one seeks to generalize the method.

Similar points can be made for the existence of other notions like obtaining of *Subformula Property*, and so on

A final TL;DR: The apparent peculiarities of **GL** : the Löb axiom, the fixed-point theorem, the well-foundedness condition , seem less to be some special quriks of provability logic that happen to make the proof work and more like they are instances of the general structure of any logic where such a method can be applied: a soundness condition on cyclic proofs, a fixed-point mechanism for resolving self-referential interpolants, and a combinatorial bound on formula complexity, etc. Hence, understanding why each step in that proof is necessary, the questions this companion has tried to address, helps one draw further conclusions and ideas for a more general programme of applicability. The hope is, this work, would have served as a useful companion along the journey through the work that started it all.

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